

ANTRIM COUNTY AP, MI, USA

WMO: 727436

Lat: 44.986N

Lon: 85.203W

Elev: 189

StdP: 99.08

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 04863

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-16.2	-23.0	0.5	-18.0	-19.9	0.7	-14.9	7.7	-3.7	7.1	-4.2	0.2	20	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	31.3	21.9	29.4	20.8	27.9	20.0	23.7	28.4	22.5	27.3	21.6	26.1	3.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.4	26.5	21.2	16.2	25.3	20.0	15.0	24.3	71.6	28.7	67.2	27.0	63.6	26.3	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.2	5.9	5.3		-25.3	34.1	4.3	1.1	-28.3	34.9	-30.8	35.6	-33.2	36.2	-36.3	37.0
			DB	-25.5	25.1	4.2	1.4	-28.5	26.1	-31.0	26.9	-33.3	27.7	-36.4	28.7
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-5.4	-5.1	-0.8	6.0	12.6	17.5	20.0	19.2	15.8	9.2	3.4	-2.1
	DBStd	10.31	5.59	5.79	6.44	5.22	4.95	4.20	3.44	3.25	4.44	4.74	4.52	4.58
	HDD10.0	2074	478	422	342	143	29	2	0	0	4	74	205	374
	HDD18.3	4131	737	655	594	372	191	64	23	28	101	285	449	633
	CDD10.0	1193	0	1	6	22	108	227	309	287	178	50	6	0
	CDD18.3	211	0	0	0	1	12	40	74	56	25	3	0	0
	CDH23.3	2373	0	0	8	22	176	467	833	578	255	33	1	0
	CDH26.7	658	0	0	1	4	48	138	244	152	64	6	0	0
Wind	WSAvg	1.9	2.2	2.1	2.1	2.3	2.0	1.6	1.4	1.5	1.6	2.0	2.3	2.3
Precipitation	PrecAvg	820	57	40	45	64	76	80	79	79	84	88	70	56
	PrecMax	924	108	65	116	95	143	153	199	133	141	199	135	117
	PrecMin	667	19	16	14	22	15	19	29	33	17	32	28	19
	PrecStd	77	23	15	22	21	29	38	42	29	32	41	31	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	12.8	22.8	27.0	31.0	32.6	33.0	32.5	31.4	27.3	20.1	12.1
		MCWB	6.9	10.0	15.2	15.8	20.1	22.7	22.7	22.5	21.4	18.9	13.9	10.4
	2%	DB	6.4	7.6	16.2	22.4	27.7	30.2	31.2	30.1	28.1	23.0	16.0	7.9
		MCWB	4.2	4.6	10.3	13.6	18.3	20.9	22.3	21.5	19.9	16.9	11.5	5.8
	5%	DB	3.7	4.8	12.2	18.9	25.0	28.0	29.1	28.0	26.3	20.6	12.9	6.1
		MCWB	2.2	2.5	8.3	11.2	16.3	19.7	21.2	20.4	18.9	15.9	9.3	4.0
	10%	DB	2.3	2.6	8.7	16.1	22.4	26.2	27.7	26.9	24.0	17.5	11.0	3.9
		MCWB	1.3	0.9	5.2	9.1	15.0	18.7	20.4	19.7	18.2	13.6	8.2	2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	10.3	15.7	17.9	21.5	24.4	25.7	25.4	23.2	20.7	14.5	10.6
		MCDB	8.2	12.3	21.7	23.3	28.0	30.1	30.5	29.7	28.2	25.5	18.4	11.9
	2%	WB	4.3	4.8	11.8	14.7	19.8	22.5	23.9	23.1	21.7	18.6	12.3	6.6
		MCDB	6.4	7.4	14.8	20.8	25.8	28.3	28.7	27.5	25.6	21.8	15.2	7.7
	5%	WB	2.5	2.7	8.4	12.2	18.2	21.3	22.8	21.9	20.5	16.0	10.1	4.2
		MCDB	3.6	4.6	11.3	17.3	22.8	26.3	27.3	26.3	24.5	19.5	12.9	5.8
	10%	WB	1.1	1.1	5.3	9.8	16.2	20.0	21.7	20.8	19.2	14.1	8.0	2.5
		MCDB	2.2	2.6	8.4	14.7	21.1	24.3	26.1	24.7	23.1	17.3	10.1	4.0
Mean Daily Temperature Range	5% DB	MDBR	6.8	8.7	10.6	12.1	13.3	13.0	13.1	12.5	12.4	10.2	7.7	6.0
		MCDBR	8.1	11.1	14.9	18.0	17.0	15.4	14.8	14.6	15.0	14.8	12.4	8.7
		MCWBR	6.8	8.5	10.0	10.0	8.4	7.5	7.1	7.2	7.6	9.1	8.8	7.5
	5% WB	MCDBR	7.5	9.8	13.4	15.5	14.4	12.8	11.8	11.5	12.2	12.3	11.3	8.0
		MCWBR	6.6	7.9	9.8	10.0	8.2	7.0	6.5	6.6	7.2	8.7	8.7	7.4
Clear-Sky Solar Irradiance	taub	0.287	0.295	0.320	0.352	0.383	0.398	0.393	0.386	0.374	0.340	0.304	0.280	
	taud	2.336	2.297	2.379	2.355	2.332	2.317	2.348	2.370	2.391	2.478	2.508	2.424	
	Ebn at Noon	825	889	915	909	886	871	870	865	845	831	809	802	
	Edh at Noon	77	97	104	116	123	126	120	113	102	82	66	65	
All-Sky Solar Radiation	RadAvg	1.20	2.12	3.55	4.57	5.56	6.06	6.11	5.20	4.03	2.37	1.30	0.89	
	RadStd	0.12	0.22	0.39	0.45	0.49	0.42	0.26	0.33	0.36	0.28	0.18	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page